Advice For Introductory Physics

Before Getting Started

What kinds of introductory physics course are there?

In the American system, there are two entry points to physics: algebra-based and calculus-based introductory courses. To start an algebra-based course (such as AP Physics 1 and 2), you need a decent grasp of algebra and trigonometry; for instance, you should know the quadratic formula and the value of $\sin 30^\circ$ off the top of your head. To start a calculus-based course (such as AP Physics C), you should at least be concurrently enrolled in a standard calculus course.

Which kind of course should I start with?

Many American physics students do both, because they realize they like physics upon taking algebra-based physics, which usually comes before their first calculus course. But there's no need to start with algebra-based physics if you already know calculus. After all, calculus was invented by Newton to make physics possible; without it, you can't derive much. Most algebra-based physics courses spend a whole hour clunkily introducing derivatives without using the word "derivative". I actually found it easier to learn calculus-based physics because I didn't have to take so much on faith.

There are a few topics specific to algebra-based courses. For instance, AP Physics 2 briefly covers thermodynamics, waves, and modern physics, which are not covered in AP Physics C. But the coverage is relatively superficial, and it'll be straightforward to fill in the gaps using a standard calculus-based introductory textbook, which covers all these topics. Some particularly good algebra-based courses (or even math-free "conceptual physics" courses) come with fun experimental demonstrations, puzzles, and real-world applications, but you can fill this in as well.

Finally, if you're halfway through an algebra-based physics course, enjoy it, and are excited to deepen your knowledge, I would recommend starting to learn calculus as soon as possible. You'll almost certainly enjoy it too, and it'll pay off many times in the future.

What should I know before I start entering physics competitions?

For people coming from a math competition background, the most important thing to remember is that physics competitions require physics knowledge. The typical American 10th grader has taken ten years of math in school and *zero* years of physics. If you're a bright student that likes math, math competitions are a fun way of extending the knowledge you've spent a decade building – you already have the foundations set, and you can easily start by learning a few extra tricks.

It's tempting to jump directly into physics competitions with the same attitude. After all, physics is just made of equations, which are math, right? If you haven't learned introductory physics already, this attitude will make you crash and burn. It typically results in people memorizing big lists of equations without being able to answer the most basic conceptual questions, and making ridiculous mistakes like confusing tension T for time T because they're the same letter. Without introductory physics under your belt, you're in the same position as a 1st grader is in math, trying to do a math competition without even knowing how to add.

Resources for Introductory Physics

How can I learn physics by myself?

You're in luck, because there are more excellent resources for learning physics independently than ever before! My most important advice is to stick to a resource that's designed as a cohesive whole, refined over time by one or two experts, and vetted by students. It will ensure that you learn the subject comprehensively, in a sensible order, with appropriate built-in practice.

When you embark on a journey like this, it can be daunting to choose among the many options; fortunately, many options can work. In particular, you can learn perfectly well from either textbooks, recorded lectures, or live lectures. The average book is clearer than the average lecture, since books often emerge from refinements of lecture notes. But live lectures can be more engaging, because the instructor presents in a room of other students paying attention. Books can drown students in too much detail, but lectures can keep students from thinking if they're too busy taking notes. Sometimes students zone out in lecture and don't remember anything at all, but students also do the same with books, mindlessly rereading while highlighting every word. Recorded lectures can be played or paused on demand, but they don't carry the deadlines that live lectures do.

The point is that, even though there's little empirical evidence that people have different "learning styles", people do vary in what gets them to effectively focus. There is no one best option, and all of them can work, so choose whichever you like most.

What are some good, free introductory courses?

- To get started with algebra-based physics, a good first goal is to pass AP Physics 1. Two good resources are the videos by Khan Academy and Flipping Physics, which have been thoroughly tested and refined by great teachers. If you'd like more structure, try the AP Physics practice exams available free online, or find an AP Physics course either online or in person nearby.

- To learn calculus, you can get started with MIT OCW's 18.01 course. You can also go through any one of the nearly identical standard calculus books on the market, such as Stewart's, which all cover everything you need and more.[1]

- Once you know basic calculus, such as derivatives and single integrals, you're ready to start calculus-based physics. My top recommendation is Yale's Fundamentals of Physics courses.[2]

[1] Mathematicians often complain that these books aren't rigorous enough, and prefer books like Spivak's. But these books are meant to train mathematicians, not physicists. Spivak is great for that purpose, but it only has a single chapter on actually performing specific integrals, and it starts with proving basic propositions like $0 < 1$ and $1 + 1 \neq 0$. If you're interested in that kind of thing, you can start reading Spivak without any prior calculus background. (Another good starting point is the Art of Problem Solving calculus book, which has a balance of proof sketches and concrete problems.) However, you won't need *any* experience with rigorous proofs to get started in physics. After all, Newton didn't care about this type of rigor when he invented calculus. When the mathematical foundations of calculus were finally set, physicists had already been using it to solve real-world problems for centuries.

[2] Unfortunately, while MIT OpenCourseWare is generally excellent, its introductory physics courses (8.01 and 8.02) aren't so good. They are broken into many dry, short fragments, with a different instructor for each lecture. Part of the reason is that MIT does not actually have introductory physics lectures; it uses an "active learning" setup where lecture is replaced with group problem solving, and students do background reading at home. Education research proves that this works better for the average student, who would just zone out in a traditional lecture. However, while I'm a fan of employing active learning in live lectures, it's not nearly as engaging when you do it by yourself at home. If you're on your own, you might as well use a book and its exercises.

What are some good introductory books at each level?

- For algebra-based physics, commonly used books are listed in AIP's survey of physics teachers. Some examples of decent books, in very roughly increasing order of difficulty, include:
  - Hewitt, *Conceptual Physics*.
  - Serway and Faughn, *Holt Physics*.
  - Serway and Vuille, *College Physics*.[3]
  - Cutnell, Johnson, Young, and Stadler, *Physics*.
  - Knight, Jones, and Field, *College Physics: A Strategic Approach*.
  - Giancoli, *Physics: Principles with Applications*.

  Judging from reviews and survey data, Hewitt is a good option for a typical high school course, while Giancoli is good for an honors high school course, such as for AP Physics 1 and 2. However, these books are all pretty similar, so you shouldn't worry if you happen to have a different one.

- For basic calculus-based physics, there are many books, such as the ones by Giancoli, Knight, Serway and Jewett, Tipler and Mosca, Young and Freedman, and Halliday, Resnick, and Walker. They all cover the same material, with nearly identical tables of contents, and they're all suitable for AP Physics C. Most of them have titles like "University Physics" or "Physics for Scientists and Engineers". They're polished and equally good, so just use whichever you can easily get. I recommend starting here if you're concurrently learning calculus.

- For more advanced calculus-based physics, I strongly recommend *Physics* (5th edition) by Halliday, Resnick, and Krane. This book is used in college honors courses, and has significantly more challenging problems. The explanations are very clear, and I know many people who have succeeded using it. I recommend starting here if you have some prior exposure, e.g. through a regular course or an AP Physics exam, and your calculus is good.

Can I learn introductory physics from websites?

To understand the answer to this question, consider how good textbooks are created. Usually, a teacher will start a course using an existing textbook. If they have the time and energy, they'll consider a wide variety of approaches, then gradually synthesize a new one for their lectures, based on their preferences; perhaps it will be more modern, more mathematically rigorous, or more intuitive than the others. Then they'll start typing up lecture notes, and once those get refined enough, they can drop the textbook and have the students read the notes directly. Over many years, students will find errors and confusing spots in the notes, which the teacher fixes up, while accruing a large bank of classroom-tested, interesting questions from the annual problem sets and exams. Finally, the teacher combines all the materials together, and a new textbook is born. All of the books and courses I recommend in this document were made this way.

[3] If you're confused about the title, note that there's substantial overlap between high school and college courses. Most people who take physics courses in college are taking conceptual or algebra-based introductory physics, which ranges in difficulty between that of a typical high school course, and AP Physics 1 and 2. Similarly, most people who take math courses take "College Math", which is about equivalent to Algebra II. These types of courses are typically taken just to fulfill a graduation requirement, and don't usually lead to further physics and math courses. Most people who end up majoring in physics start in calculus-based introductory physics in college.

There are two active ingredients in the process. The first is the students, who act as dedicated test-readers, pushing the teacher to improve their materials year after year. Books that aren't student tested tend to be plagued with issues, such as constant typos, trivial or nonsensical problems, huge jumps in difficulty, and crucial omissions. The second is the teacher's continued attention, which ensures global consistency, and deep expertise. After all, to write a good book, the teacher must know far more than what is actually in the book. This lets them identify the big picture, understand problem solving strategies, create new problems, and see the limitations of the usual formulas.

Many websites for learning are lacking these ingredients. For example, Brilliant and Expii have lots of neat problems, but their physics curricula aren't developed in a complete and logical manner. The problems have wildly different notation, conventions, and difficulty, and units tend not to be self-contained, often requiring knowledge from later units. Recently, many vibecoded websites have appeared which look professional, but have the same problem; it is clear the author simply asked AI to generate dozens of questions without regard to whether they fit together into a useful whole.

Wikipedia has a lot of useful information collected by dedicated editors, but because of its scale, it also has many inconsistencies and mistakes. If you ever get confused reading it, e.g. if two definitions don't seem to be compatible, or if a step in a derivation doesn't seem right, you should never, ever try to resolve it by opening up twenty Wikipedia tabs. The answer is simply not going to be there, and you'll just magnify your confusion.

Popular videos tend to be even worse. On YouTube, there are millions of videos where some guy explains a physics topic off the top of his head. The problem is that most of these people have only learned the basics the previous day, often by skimming Wikipedia. Because they're just talking off the top of their heads, their videos tend to be vague, inaccurate, stuffed with filler, and way too long, since longer videos pay more. Sometimes students instinctively try to fix this by cranking the video speed up to 3x, which is almost always a mistake. If you ever get the urge to do this, it means the video carries too little new information to be worth watching.

More generally, while it's fun to discuss physics with other students, you generally shouldn't invest serious effort into learning from resources written by recent students. Again, these have the problem of lacking deeper expertise and student-testing.

When I was a kid, I followed the procedure for information gathering taught to me in public school: Google the term, open the top ten links, and then open all the links in those pages. The typical result was that I'd get lost all day on an issue that should have taken five minutes, wondering in despair why somebody didn't just organize the material consistently. Only later did I realize that this is literally what books and courses are! In fact, when YouTube and Wikipedia explanations are complete and reliable, they've usually been copied from a book. If the book is a source of illumination, these secondary resources are just the shadows it casts. The internet is the world's greatest source of amusing pictures and videos, but it's not great for anything deep. The sum total of all physics discussion online contains less content than a few stacks of good books.

Can I learn introductory physics from AI?

As of late 2026, leading AIs can reliably explain anything in introductory physics. However, I haven't yet seen any compelling examples of somebody teaching themselves a large, difficult subject from scratch using AI. The first issue is that AI lacks detailed long-term memory, so it can't fully track what you know and don't know. Second, AI responds with walls of text, while lectures and books can fluidly intersperse diagrams and demonstrations. Third, AI generally responds to what you want in the moment, rather than challenging you with what's best for long-term growth, such as a tough problem you'll get stuck on but remember forever. Fourth, since it's seen everything

before, AI seems poorly calibrated on how difficult problems feel for new learners.[4] (Professors can have this problem too, but they also get regular teaching practice.)

Of course, these problems can be fixed with the right prompt and harness, but if you don't know the subject yet, it's hard to know if your setup is working well. The best case for using AI is that it can help personalize a course to your tastes, but I don't think that applies for introductory physics, which is already highly streamlined. (There's no universe in which your interests would mean you should learn about energy but not momentum!) So I don't see much potential gain from AI yet. This may change in the future, if companies build dedicated AI tutors synced to good curricula.

You might also wonder if it's okay to ask AI about points you're confused about. Personally, I don't think there's anything inherently wrong with doing this. But it's just like reading problem solutions: you can lose out on deeper learning if you turn to it too quickly.

Why are there so many similar introductory physics textbooks?

Like most physical things in America, introductory physics textbooks were built in the postwar period. After Sputnik, concerned scientists and policymakers made a societal push for STEM education. This gave rise to many great books, such as the original Halliday and Resnick, and the Feynman lectures. Halliday and Resnick was the dominant calculus-based physics textbook in the 1960s, and things haven't changed much since then. Newer textbooks are direct descendants, sometimes taking topics out, but rarely adding new topics in. For example, Halliday and Resnick itself split into two versions, *Physics* and *Fundamentals of Physics*, by Halliday, Resnick, and Walker. The latter is essentially just *Physics*, but with the most advanced parts of each chapter removed.

You might also notice that these books come in many editions, and that the latest edition is much more expensive than the rest. For example, Serway and Jewett is on its 11th edition, while Young and Freedman is on its 16th. However, you shouldn't worry if you can't afford the latest edition, or if you happen to already have an earlier edition. The core introductory physics curriculum hasn't changed for decades; the real purpose of the endless editions is to keep money steadily flowing in for the publishers. To make a new edition, they fix a few typos, then randomly rearrange the ordering of the problems and the numbers inside, and lobby university administrators to make their instructors have their students buy it.[5] In other words, the main purpose of new editions is to make money by preventing students from buying cheap used copies.[6]

Of course, most of the time, you should prefer the latest edition of a book. They tend to have fewer mistakes, and sometimes better content; for example, the latest (5th) edition of Halliday, Resnick, and Krane has many very useful multiple choice questions, and some extra tricky problems. But textbooks with over 10 editions are often just bloated. Some run to almost 2,000 pages, weighing as much as a newborn child, because they're crammed full of redundant plug-and-chug problems

[4] One vibecoded website with AI-generated difficulty ratings ranks APhO problems below $F = ma$ exam problems.

[5] They also force students to buy online homework assignments, infamous for their clunky design and 1990s-level programming. Their automated system will mark you wrong for typing 5.0 instead of 5.00. It's precisely the sort of thing that takes the joy out of physics, and turns it into dull accountancy.

[6] If even a used copy is too expensive, you can consider buying "international" editions of textbooks, which are sometimes marketed by publishers as "global", "Indian", or "South Asian" versions. (For example, one of the international versions of Halliday, Resnick, and Walker's *Fundamentals of Physics* is called *Principles of Physics*.) These books are printed on thin paper with a flimsy softcover binding, and cost about 80% less. Usually they have comparable content, but various things like the table of contents or the index are made worse on purpose, so that people have a reason to buy the full-price version. I used to try saving money in college by buying Indian editions, but when I got Griffiths' electromagnetism textbook, the whole chapter on magnetism was missing. These textbooks are still good enough to learn from, but if you have a choice you should prefer the original version.

and glossy but misleading pictures. This is all just to say that while physics is certainly real, there are a lot of things about the physics education system that are downright silly. Fortunately, if you're self-studying, you can avoid these problems entirely.

How about alternative books?

Some folks have tried to address high textbook costs by writing free textbooks available online. The most popular ones are on OpenStax, which has released nearly 100 textbooks in the past few years. Unfortunately, while their mission is a great one, these proto-books aren't ready to replace polished, published books. The most rigorous OpenStax book, titled "University Physics", is pitched at a significantly lower level than the calculus-based books listed above. Each chapter has about half as much content, and almost zero nontrivial problems. Derivations often lack consistent notation or the cleanest possible approach, and problems are often formulated vaguely or incorrectly. This is likely because the book was cobbled together by two dozen authors.

There are also many supplemental books made for AP exam preparation, such as the Princeton Review, Barron's, and 5 Steps to a 5 series. I generally don't recommend them. They tend to have much higher average review ratings than real textbooks, but that's because the reviews are left by students who want to cram to pass, not learn. They are designed to get you through the simplest questions with the least possible mental effort, and as such, don't really explain how or why anything works. Not only does this suck all the joy out of learning, it'll leave you unable to answer any question deeper than a one-step plug-and-chug. They may be okay for a very quick first exposure, but you'll want to upgrade to something better quite soon.

Physics Competitions

How does the American physics competition system work?

Every year in America, about 175,000 students take AP Physics 1 (algebra-based mechanics), and 65,000 and 30,000 take the AP Physics C mechanics and electromagnetism tests, respectively. Not all of them know about or are interested in physics competitions. But annually, about 6,000 students take the $F = ma$ exam, from which 400 students qualify for the US Physics Olympiad, 20 qualify for the training camp, and 5 are selected to travel to the International Physics Olympiad.

How long will it take to qualify for USAPhO/camp/IPhO?

This varies depending on the person and their motivation, but here's my timeline.

- 9th grade: I took a standard pre-calculus course in school and didn't know or learn any physics.
- 9th grade summer: I don't recall learning anything. I grinded a lot on RuneScape, with occasional breaks to practice for math competitions. (This didn't help for physics competitions at all, besides making me a bit faster at algebra.)
- 10th grade: I took algebra-based introductory physics in school, along with AP Calculus, with great teachers in both. I didn't prep for competitions, but I asked a lot of questions in class, thought carefully about the intuition behind the equations, and occasionally skimmed the mediocre *Holt Physics* book given. I just barely qualified for the USAPhO, and scored almost zero on it. But I found that experience very motivating, since it showed me that physics was full of cool problems, which took a lot more than just plugging numbers into a formula sheet.

- 10th grade spring/summer: I self-studied calculus-based physics by reading the absolutely terrible Barron's AP Physics C prep book and randomly googling whenever I got confused. This took roughly 150 hours of work. Some of this was done while avoiding MOP homework.

- 11th grade: I read the wonderful textbook by Halliday, Resnick, and Krane, thoroughly understanding about one chapter per week, and mixing in past $F = ma$ exams in January and past USAPhOs in the spring. I didn't use any other textbooks, or attend any prep courses or camps, and my school had no dedicated physics club. I just self-studied roughly 10 hours a week, for about 250 hours in total. (It wasn't hard to find the time, since I took a minimal courseload, with a single AP class.) That year I qualified for camp, worked through ten past IPhOs as recommended by the camp coaches, and got an IPhO gold medal.

The point is that you don't need a decade of study or a ton of prep programs to succeed. And this isn't just my experience. When students on our IPhO team describe their journey, they usually say something very similar. After learning introductory physics, usually in high school, they prepare for physics competitions for a year, or two if they have lots of other things going on. Most camp qualifiers don't take any prep courses at all, but those that do only take one. All of them think about physics regularly, but none of them thought of their training as a source of suffering. Training is tough but fun, because physics is fun, and the best sources are no secret.[7]

Do you need Math Olympiad background to do well in the Physics Olympiad?

No. Having done both, there's very little overlap in the skills you need. The ideas that come up in combinatorics and number theory problems are almost never useful in physics. As for geometry, there's never been a USAPhO or IPhO question that required more than the most basic properties of conic sections. Doing algebra problems might make you faster at simplifying complicated expressions, but those are rare in physics competitions too. Indeed, most of the members of the U.S. Physics Team have never even taken the USAMO, and the vast majority have not spent any time preparing for it. To do well at physics, you study physics, not triangle centers.

There are two reasons people might think the Olympiads are related. First, as I mentioned above, Americans only start learning physics in school ten years after starting math, so people who like physics competitions will probably have tried math competitions first. Second, in both cases a student succeeds by teaching themselves a well-defined body of knowledge by struggling through hard problems under time pressure, so developing the grit to do that for math can make it easier to do the same for physics. But math competitions are absolutely not a prerequisite: you can jump into physics even if you've never done a math competition.

Is prep program X, book Y, or course Z enough for USAPhO?

Any decent calculus-based physics course, book, or prep program is "enough", in the sense that they'll all cover everything you need. But it's up to you to turn that coverage into understanding!

[7] Even bad sources can work. Recently, our IPhO teams have had students who studied with only 70 year old, $10 Dover books, or were homeschooled with no physics teacher or tutor, or who used only the OpenStax physics book, or who used nothing but Wikipedia! I don't mention this to encourage you to do the same – they all would have had an easier time if they had used better books from the start. But new students often have the impression that they're doomed unless they use twenty books or the most expensive prep programs. In reality, everybody's journey is different. The only common factor is thinking deeply about physics, and enjoying it.

What makes a competition prep program effective?

The main thing that makes a prep program effective is the student: if they aren't interested and engaged, then any program will be useless. This is obvious if you just look at the numbers. Suppose an unmotivated student is dragged to a 1.5 hour class every week for eight weeks, then grudgingly spends an hour a week on the homework. That only adds up to 20 low-quality hours of learning. If practice stops entirely once the class ends, the knowledge will be quickly forgotten. As such, there are a lot of people dragged to prep programs year after year, who never get very far. These programs are very popular in certain areas of the US, but in many cases seem to be a waste of time and money. Worse, when people are *forced* to do physics, they often end up hating it, which is exactly the opposite of the original purpose of the Olympiad.

Compare this to what I listed above: 400 hours accumulated over a single year. Objectively, that isn't a lot of time; people could easily spend longer than that on a single high-school course if it's loaded with busywork. But these hours were focused ones, and they were spaced out regularly. I didn't need to cram, because I'd been immersed in physics the whole time.

I sometimes see parents spend more energy dragging their kid through prep classes and books than their kids spend actually thinking about physics. Sometimes they even solve the problems for their kids! Parental involvement is like salt. A pinch can enhance a dish, but too much overwhelms the taste, and adding more makes it inedible. If a kid is fundamentally uninterested in something, they should be free to do something else. There are many things to do in this world besides physics.

But doesn't everybody go to prep programs?

Prep programs design their advertising precisely to make students worried about this. For example, Art of Problem Solving's PhysicsWOOT program regularly boasts how many US Physics Team members are in their course, but this is simply because the team members from the previous year get enrolled for free. Similarly, Tang Academy counts all of its alumni in its statistics, even those who only enrolled in introductory physics years ago.[8]

In reality, despite their advertising, no prep courses actually operate at the level that would be relevant for the US Physics Team camp, because there would be too few students to make the class worth teaching. Instead, the main benefit of prep courses is in providing a friendly, encouraging entrance to Olympiad physics, at the level of the $F = ma$ exam or early USAPhO. But that doesn't mean they're *required*. Prep programs that describe the Olympiad as a bag of isolated tricks don't actually work, while those that take the time to build a solid foundation are just the same as any decent course or book. There's no proprietary secret to learning physics. You just have to think.

If it's that simple, why doesn't everybody ace the USAPhO?

Many people talk about preparing for the Olympiad these days, but from my experience, out of every 100 people who ask, only 25 even bother opening a good introductory physics book, only 5 make it past the first few chapters, and only 1 actually deeply understands the whole thing. Really studying physics takes time and consistent effort, which filters out people who aren't interested in the subject. So at least in America, if you've thoroughly studied a good book, you're competing

[8]The advertising works extremely well on Asian parents, too. Quite a lot of US Physics Team camp members were sent to these programs by their parents, even when their level was already above what the program taught.

against less than a hundred other people.[9]

I can't solve problems involving pulleys. Is there a dedicated book about pulleys?

There's no such thing. Beginners tend to classify physics problems by their superficial features, so that they'll often say, e.g. that they "can" do inclined plane problems but "can't" do pulley problems. Or they may think, after training on the many similar pulley questions in Morin, that every problem involving a pulley is solved with the idea of "conservation of string".

Once you understand physics more deeply, you'll see that all of those classes of problems are governed by the same basic principles, i.e. Newton's laws, plus energy and momentum conservation. In general, if you find a problem unusually challenging, you should drill down and figure out what you're missing about the underlying principles, then review it in any decent textbook.

If I know X, can I read book Z or should I read book Y first?

It depends on the context, but the answer that always works is to just try it and see. If you have a lot of trouble with the easiest exercises in a given book, then you should use an easier book. If you find the easier exercises easy, then you should focus on the harder exercises. And if you feel you're not learning anything new from even the hardest exercises, then you should use a harder book.

There's an essential core to physics, shared by all good books. As long as you understand the resource you're using, and you're learning new things, you're on the right track.

Do I really have to learn X if I want to win competitions?

For almost any value of X, the answer is "probably not", but if you ask this kind of question constantly, you won't do well anyway. Stop and find a different extracurricular, one where you're excited to do more rather than bargaining to do less.

Jeez, okay, but can I qualify for USAPhO without knowing calculus?

Every problem on the $F = ma$ exam should be solvable without calculus, but most students who pass the exam know calculus-based physics. The reason is that it's hard to derive many simple physics equations without using calculus. And if you don't know how the equations are derived, you might only see them as a disconnected pile of results instead of an interconnected web of ideas. This penalizes you on the $F = ma$ exam, where many questions require the test taker to think very carefully about which equations apply and why. It's certainly not impossible to pass without calculus, but you're going to have to put in the time to build a solid conceptual understanding either way – and it might end up taking *longer* if you try to do it without calculus.

And do I need to learn advanced topics, like linear algebra or quantum mechanics?

Students often think this is necessary for three reasons: (1) Olympiad questions are often related to advanced topics, (2) successful Olympiad participants often know a lot of advanced physics, (3) students preparing for math competitions often read big stacks of Art of Problem Solving books.

[9] For over a decade, students have said "that advice might have worked last year, but people are just too cracked this year!" There's certainly a difficulty gap between the 2024 and 2008 USAPhOs, but everything happens gradually. From my experience teaching at the camp since 2015, each year the top students know 3% more, work 2% faster, and get 1% cleverer. There are never any big jumps, and there's no point in worrying about hypothetical competitors.

All of these are misleading. The Olympiad syllabus is centered around introductory physics, and questions are carefully designed to be solvable without any advanced knowledge. (In fact, trying to use more advanced tools often results in a slower solution!) Successful participants often know a lot of physics not because it's particularly helpful for competitions, but simply because they like learning things. And a single introductory physics textbook is a lot denser than an Art of Problem Solving book, because it has to build a whole subject from the ground up. It's great to explore more advanced topics, but at the same time, you shouldn't feel obligated to.

Does my comprehensive multi-year study plan look good?

Students often devise detailed plans, involving reading huge stacks of textbooks and working through thousands of problems. After over a decade of experience, I haven't seen a single person actually follow through on such a plan. Without experience in physics, you can't estimate how long each step will take you; moreover, your interests and priorities will naturally change over time. Even I fell into this trap when I started college. I spent a day making a four-year plan in the first semester, only to completely scrap it when the second semester started. Then I made a new plan, and scrapped it when the third semester started.

There are two reasons overplanning happens. First, it's a productive-feeling form of fantasy. It's fun to think about how much you might learn in the future, and it takes less mental effort than actually learning something now. Second, it helps assuage anxiety. It feels good to think that everything is accounted for, though in practice, things will never turn out the way you expect. In any case, a little overplanning is harmless, as long as you remember to actually take the first step.

Can a professional physicist get a gold medal at the IPhO?

Though most mathematicians can't get a gold medal at the IMO, a well-rounded physicist who regularly teaches *should* be able to get a gold medal at the IPhO. That's because math competitions focus on niche topics like Euclidean geometry which rarely come up in higher mathematics, but can be scaled up to arbitrary difficulty. By contrast, physics competitions were invented to spark interest in physics, and to teach students to think like physicists. Climbing to IPhO level will take you on a tour through some of its greatest ideas, from the ones that Newton pioneered to those behind recent Nobel prizes. Everything you'll learn is key to our understanding of nature.

Tips for Self-Study

How can I meet other students?

The most active English-language community for Olympiad physics is this Discord server.

Arrr! How can I preview textbooks and papers before buying them legitimately?

For books, try the "shadow libraries". For papers, try SciHub.

How can I learn introductory physics most effectively?

Here are some basic principles that almost everyone agrees on.

- Actively engage. When you read about a new physical idea, turn it over in your head. Ask yourself where you've seen the idea at work in the real world. Look at its logical development – what assumptions do you need to get from one equation to another? Get a feel for how each equation behaves as the variables vary. Take limiting cases of them, relating them to ideas you already know, or try to go beyond, seeing where they might fail. Try to reconstruct the idea, in a way that makes it intuitive. Do practice problems, or make your own. Many students don't do any of these things, but claim they "understand the concepts" because it "all makes sense". They are experiencing the "illusion of explanatory depth", mistaking familiarity for true understanding. If you're actually thinking about the material, you *should* occasionally get confused; learning comes from noticing and resolving such confusions.

- Avoid skipping around. Sometimes, students skip straight to the exercises, figuring that they can "learn by doing." Then, when they get stuck, they haphazardly skim the text for equations which seem to contain the right letters. (This is also a common problem for study groups working on problem sets.) This procedure tends to make one-minute exercises take an hour, and doesn't work for nontrivial exercises at all. When you're starting out, it's most efficient to spend a comparable amount of time learning theory (which includes both reading and thinking) and solving concrete problems afterward. And theory should be learned linearly, in the same order as the book's chapters, to avoid unnecessary confusion and backtracking.

- The best way to remember something long-term is spaced repetition: apply the idea the moment you learn it, then reencounter and reuse it regularly. Good physics books and courses will automatically make you do this, as long as you work steadily and linearly through them, solving a good number of problems.

- Do practice problems just above your current level. Trivial exercises alone won't make you ready for harder ones, and if you do too many you'll probably get bored, zone out, and start making dumb mistakes. On the other hand, if you skip to very hard problems, you might spend long stretches of time making zero progress, or trick yourself into thinking you see the answer without understanding the subtleties at play. Aim for a difficulty level where you can work out most, but not all, problems completely by yourself. Your practice should feel effortful, deliberate, and possibly a little uncomfortable.

- Don't fool yourself by relying on solutions. A solution is useless if you haven't already given a problem a good try first. Many students end up exclusively reading solutions without being able to work out anything for themselves, but feel confident because the solutions "make sense". ("I got it wrong at first, but now that I've read the solution I know it's easy!") When the inevitable happens, students often say they "froze up", "blanked", or "don't test well". But the truth is that they didn't forget anything – they never developed the skills in the first place.

- Get feedback. When you finish doing a set of practice problems, reflect on how they went, and if you weren't able to do some of them, figure out the crucial steps you were missing. This is especially important if you're self-studying, as there won't be many other ways to get timely, objective feedback, which is absolutely necessary for learning.

- But don't get feedback *too* fast. It's tempting to run to the solution manual (or to AI) the moment you get an idea, to see if you're right. But generating ideas is only part of the problem solving process; you also need to implement the ideas, check whether they're sound, search for alternative ideas, and (most subtly of all) evaluate when you can stop searching. These skills

are all crucial for competitions, research, and life in general. You should only check a solution once you've gotten totally stuck, or carefully thought and committed to a final answer.

- Make sure your studying is healthy. Long cram sessions aren't effective. Take regular breaks instead of sitting motionless for hours. Sleep enough to feel rested, drink water, eat food, and generally obey common sense. Studying when your brain or body is tired is only useful for mindless tasks like cramming things into short-term memory, the opposite of what you need.

  On a related note, I recommend spending at least 1% of your time (15 minutes per day) exercising. If you don't currently exercise, even picking up jogging or simple bodyweight exercises will make everything in life easier. Even if you only care about studying, the time spent exercising will pay off for itself by improving your energy levels. (It even seems to make people better at experimental physics exams!) Back when I was training for the IPhO at home, I would do a few pullups for inspiration whenever I got stuck.

A decent heuristic to check if a learning plan makes sense is to ask if it would make sense for learning to play an instrument or a sport. Nobody has ever become a great musician just by watching others perform, or just by endlessly practicing scales and arpeggios. Nobody's ever given a great piano performance by "cramming" a piece from scratch for 10 hours the previous night.

An athlete on their country's youth soccer team didn't start by getting in contact with the national team coach. Instead, they probably started by playing lots of casual games with their friends, gradually building up their skills while having fun. As they got better, the stakes were gradually raised, until they ended up going to tournaments and doing daily deliberate practice with coaches. In both sports and physics, long-term motivation is maintained by small, consistent wins. The athlete gets through a 6 AM practice session by focusing hard on improving their fitness and technique. The physics student leaves a study session excited by what they've learned about the world. In both cases, you have to enjoy the game.

Do I need fancy equipment to study well?

As long as you're actually thinking about the ideas and working through problems, the specific way you do it doesn't really matter. Old people often insist that the one *true* way to learn, as intended by millions of years of evolution, is to write full English sentences in cursive with a fountain pen in a leather-bound notebook. But I type my notes in bulleted lists, and I have smart friends who like to debate with others, or walk in circles mumbling to themselves, or just stare into space. Use whichever method you like best, and it'll work as long as it keeps you engaged with the ideas.

In fact, I generally advise against having too complicated of a study method, as it can distract you from the physics. For example, there are many online advocates of building elaborately linked "mind maps", such as Zettelkasten, but they often end up spending more time maintaining the links than actually having new thoughts. Another trap is to try to remember everything, e.g. by highlighting every word of a textbook, or writing notes as long as the textbook itself. This doesn't help because understanding requires compressing information, which in turn requires thinking.

Finally, I don't recommend using mnemonics or flashcards for learning physics. Flashcards are good for things that are broad, shallow, and repetitive, like learning vocabulary words in a foreign language. But in Olympiad physics there isn't that much to memorize, and solving new problems requires *actually thinking*, instead of kneejerk reacting by rewriting standard equations. Mnemonics are useful in other fields because they can give some artificial meaning to something totally random,

and people are best at remembering meaningful stories. But physics equations aren't random: they actually make sense, and if you understand why they make sense, you'll easily remember them.

Do I really need to solve problems? / Do I really need to understand theory?

Some students say "I don't need to do any calculations, because I really understand the concepts!" Others say "Who cares why that works – I got the right answer this time, didn't I?" But both are misguided, because to solve nontrivial problems you'll need to be comfortable with both sides. A well-designed exam will force you to consider new situations, where you'll need solid understanding to know what steps make sense.

Every great physicist has been strong at both conceptual understanding and problem solving. Popular science books portray Newton as a daydreamer hit by an apple, but as a young man, he filled notebooks with elaborate calculations for fun. Maxwell created a new way of thinking about electromagnetism, but at college he grinded for years on "problems made express to bore me."

There are particularly many myths about Einstein. People say he was a patent clerk who figured out special relativity in a stroke of intuitive inspiration. But that story is deeply misleading: Einstein taught himself calculus at age 14, aced the entrance exams for a top university, and wrote his *annus mirabilis* papers while finishing his PhD at a top graduate program. (He only worked at the patent office for a short time while waiting for a professorship to open up.) People also say he came up with general relativity by having his "happiest thought" in 1907. But in reality, he struggled for over 6 years to turn that thought into a real theory, which required intense calculations.

How should I self-study from Halliday, Resnick, and Krane?

A good pace is about one chapter per week, or three chapters per two weeks. If you're self-studying, you should continuously test your knowledge to avoid gaps in understanding. While reading a chapter, spend at least as long thinking about its contents as you do physically reading the words. Afterward, spend at least a moment thinking about every conceptual question, as many of them help you connect the theory to the real world, and some are surprisingly deep. I also recommend doing all of the multiple choice questions, since they're good preparation for the $F = ma$ exam. But it's not worth doing all of the exercises, since many are just plugging numbers into standard formulas. Skim them to see if you know how to do them, and try a few to check.

On the other hand, I recommend reading all of the problems carefully, and doing at least half of them, depending on which strike your interest. Note that while some of the problems aren't too different from exercises, a few (such as 4.12, 5.11, 7.12, and 17.8) are much harder. You shouldn't get discouraged if you get stuck on them, but you shouldn't ignore them either, as these tougher problems illustrate key ideas that are very important on the Olympiad.

Answers to odd-numbered exercises and problems are at the end of the book. There's an official solution manual online, but its explanations are brief, and sometimes sloppy. (It's also not necessary: when I was preparing, I didn't even know there was a solution manual!) There are no official answers to the multiple choice questions, but you can find my answers for the first 17 chapters here.

Tips for Competition Preparation

How should I prepare for the $F = ma$ exam?

The main ingredient for success is a solid understanding of mechanics, which you can pick up from any calculus-based physics textbook. I particularly recommend the first 17 chapters of Halliday, Resnick, and Krane (5th edition), since it has plenty of tricky multiple choice questions. Once you're done, you should also prepare for the quirks of the $F = ma$ exam, but don't get the priority flipped: exam preparation should take a couple dozen hours at most, while learning the foundations takes hundreds of hours. And the foundations come first.

Anyway, the $F = ma$ exam throws tricky multiple choice questions at you under extreme time pressure, and the best way to prepare for that is to train on similar problems under timed conditions. There are over twenty past $F = ma$ exams publicly available, which gradually increase in difficulty over time. If you want an even gentler start, you can try the ten $F = ma$ exams from before 2007, though many of their questions are quite routine.

After a practice exam, you should immediately check against the answer key, and understand how to solve any question you missed. (Earlier $F = ma$ exams don't come with detailed solutions. For 2011 through 2019, you can use the solutions in the book by Kisacanin and Zhang, which are distributed for free on the AAPT website. From 2018 onward, there are also detailed official solutions.) Another excellent resource is Morin's *Problems and Solutions in Introductory Mechanics*, which contains a lot of multiple choice questions, with explanations, at about the right level.

If you run out of problems, you could also try past PhysicsBowl questions, the CAP prize exam, the first round of the British Physics Olympiad, or the Hong Kong Physics Olympiad. However, these competitions are generally more straightforward, and some contain non-mechanics questions. The existing $F = ma$ exams should be more than enough if you use them thoughtfully.

How can I stop losing lots of points on the $F = ma$ exam due to silly mistakes?

First, understand that "silly mistakes" are likely not just due to issues with basic algebra; instead, they are much more often due to gaps in your understanding of mechanics.

- Check if the results make sense. For example, if you manipulate $F = ma$ into $a = m/F$ you'll conclude that an isolated object has infinite acceleration, which makes no sense. More generally, you can often simplify a problem by taking limiting cases, i.e. by considering when some of its parameters go to zero, infinity, or become equal to each other. Check that your answers make sense in as many limiting cases as possible. If possible, visualize your answer happening with objects in the real world, and see if it fits your intuition.

- Check the *ingredients* of a calculation. If you make a mistake in a calculation, and you check it by just reading it again and running through all the steps in your head, in the same order, you'll probably skip right over the mistake. Instead, try to check in a way that separates the problem's essential parts. For example, I like to do a "starting point" check to see if every ingredient going into the derivation is right. Then, for the derivation itself, I do a "no number" check where I ignore all the numbers, and just make sure that the variables are showing up correctly, i.e. that something that should be in a numerator doesn't end up in a denominator. Finally I do a "number only" check where I only track things like factors of 2. In each check I have to pay attention to only a few things at once, which makes it easier to catch errors.

- Organize your work as linearly as possible, keeping work from a single problem together. If you realize something you wrote is wrong, make sure you clearly mark that in some way.
- Understand the reason a problem is tricky. If you think a problem can be solved in five seconds by directly applying a single standard formula, then it's worth thinking for a whole minute about whether that formula actually applies.

Any advice for taking the USAPhO exam?

After thoroughly learning calculus-based introductory physics, you can mix in previous exam problems for practice, and dive deeper into specific subfields using the resources listed in my second advice file. Here are some tips for the exam itself:

- Problems gradually increase in difficulty in time, so it's best to try USAPhOs in roughly chronological order, starting in 2007. For an even gentler start, you can try pre-2007 USAPhOs, or even quarterfinal exams, which used to go between the USAPhO and $F = ma$ exam.
- It's good to practice under realistic conditions. When doing a USAPhO problem, work on it uninterrupted for at least the full time limit (i.e. at least 30 minutes for recent USAPhO problems), and write a solution as you go, boxing a final answer. Good problems are scarce, so you should never waste one by turning to the official solution before giving it a real try.
- The USAPhO has generous partial credit, as you can see from a past rubric here. To get the most of it, you should write your solution clearly. The logic should flow linearly down the page, and your handwriting should be legible, especially for your final answers. You don't have to write full sentences, and you can get full credit without writing a single English word, but it's good to add short explanatory phrases (such as "by conservation of energy" or "by solving the above two equations"). If you make a mistake on one part that propagates to a later part, you generally won't get double penalized, unless your mistake changes the problem qualitatively (e.g. by violating dimensional analysis or limiting cases).
- Anything covered in a standard introductory physics textbook can be used without derivation or justification. (More advanced tools, such as Lagrangian mechanics, can also be freely used, but usually won't help. For instance, 2019 B3 can in principle be solved with Lagrangians, but in practice it is much easier to just think carefully about the definition of acceleration.) On the other hand, 2022 B2 centers around *deriving* the Lorentz invariance of electric charge, $Q' = Q$, using a relativistic thought experiment; in that case you clearly can't get full credit by ignoring its subparts and just writing $Q' = Q$. So the general rule is that you can use any result, unless the problem is explicitly about deriving that result.
- In high school, you might have been made to follow silly rules to avoid losing points. For example, you might have always had to draw a free body diagram, even in cases where it wasn't necessary or didn't help. Or you might have had to write equations in a particular format, such as $x_f = x_i + v_0(t_f - t_i) + a(t_f - t_i)^2/2$ in a case where $d = at^2/2$ would have sufficed. These rules are crutches, enforced by teachers because they make grading easier, and help most students reliably solve plug-and-chug problems. They don't apply on the USAPhO.
- In some high schools, you can get a lot of partial credit by just doing a bunch of meaningless work. For example, if the problem asks for a final number, you can make up a random wrong

expression and plug made-up numbers into it. Or, if you have no idea what to do, you can just write down every random formula you know. This is worth essentially zero points on the USAPhO. Partial credit points reflect the actual progress you make in solving the question.

- As for simplifying your answers, use common sense. If the answer is $\sin\theta$ and you write $\sqrt{(1-\cos^2(2\theta))/(4\cot^2\theta\sin^2\theta)}$, you're going to lose a few points. If the answer is $2\sqrt{2}$ and you write $\sqrt{8}$, it's fine. There are no secret weird rules. Just simplify so that the structure of your answer is clear. That will also help you tell if your answer is right.

- You should also reserve a few entire USAPhO exams to take in one sitting, to practice time management during a full exam. I recommend reading all of the problems at the start of the exam, and beginning with whichever looks most approachable. It's important to avoid spending too long on any one problem, as some can be much harder than others. If you get stuck, try moving to a different question, or going back to check your work on an earlier question.

- The USAPhO routinely covers completely new ideas (e.g. liquid-gas phase transitions in 2015, op amps in 2016, entropy conservation in 2017, diodes in 2018, rotation with a changing pivot in 2019, nonideal gases in 2020, convection in 2021, and bending moments in 2022). You shouldn't be discouraged if they look unfamiliar. These questions are designed to test your ability to learn and use new concepts, as would happen on the IPhO, and they always contain enough information to solve without special knowledge. Be ready to adapt!

- According to past statistics, the median student gets about 20% of the points, and almost no students get above 75%. So in the modern USAPhO with its 6 equally weighted questions, a *very* rough guide is that the honorable mention cutoff corresponds to 0.5-1 questions, bronze to 1-2, silver to 2-3, gold to 3-4, and slightly more than that qualifies for camp. (Of course, the cutoffs vary from year to year depending on the difficulty, and you can also accumulate points through partial credit.) If you passed the $F = ma$ exam and know calculus, it's not too hard to get an honorable mention by making an honest effort on the mechanics problems, and if you want to go to camp, you have to solve most of the exam, but not the whole thing.

- Don't give up if the problems seem hard, or get complacent if problems seem easy, or obsess over exactly where the cutoffs are. Difficulty varies from year to year. Almost everybody comes in with a similar introductory physics background, so if you feel a test is unusually hard or easy, then others will too. During the exam, you should spend absolutely zero time worrying about how others are doing, and just concentrate on doing the best you can.

Escaping School

Do I need to know the weird trivia I learned in middle school?

The standard American public school science curriculum has a lot of things that don't really make sense. For example, we are told to remember that there are exactly 3 kinds of lever, 4 ways to write the equation of a line, 5 states of matter, 6 kinds of simple machine, and 7 steps in the official Scientific Method. I was told to round numbers to the closest digit, unless it was a 5, in which case one should round to an even digit, unless the number was negative, in which case one should round to an odd digit. In some schools, students must remember that the pound is really a unit of mass; the unit of weight is called the pound-force. In other schools, they must remember that the

pound is really a unit of weight; the unit of mass is called the pound-mass. In some schools, they do multiplication and division from left to right, so that $1/2 \times 3 = 3/2$. But in other schools, they do multiplication before division, so that $1/2 \times 3 = 1/6$, and in others, division comes first.

When I was a kid, I thought this minutia was incredibly boring. It turned me off science, which seemed to boil down to the drawing of arbitrary distinctions and the memorization of arbitrary rules.[10] Thankfully, none of this trivia matters for the Olympiad, or physics in general. It's cargo cult learning, which vaguely looks like science, but in reality just keeps you busy.

But if it's really that bad, why are millions of kids subjected to it? And why do tens of thousands of teachers repeat it year after year? The underlying problem for the teacher is that genuine, interesting problems are often too difficult to assign. Covering minutia is a convenient alternative, because most students can be trained to do it, and an infinite number of quiz problems on it can be easily generated and graded. That's why, when the physics education researcher Edward Redish once asked his students what the most important equation in mechanics was, the most common response was not $F = ma$ or energy conservation, but $d = at^2/2$.

In math-heavy subjects, such as physics, there often isn't enough genuine minutia to fill a whole course. So curriculum designers compensate by making up *fake* minutia that no professional actually needs, such as the ten different mechanical advantage formulas, or the amazingly complex rules for rounding. Often, the rules you're supposed to memorize don't even agree from school to school, and the reason is that they truly don't matter.[11] No puzzle in physics has ever hinged on whether the One True Order of Operations was PEMDAS or PEDMAS, even though people never seem to tire of debating it on social media. If you're like how I was as a kid, you'll want to ignore this noise altogether, but unfortunately grades are still quite important at this stage in your life. My advice is to learn the minutia just well enough to maintain decent grades. Treat schoolwork as a day job and save your energy for deeper things.

Of course, some of the arbitrary-looking stuff you learn in school actually does turn out to be important. For example, you'll spend a lot of time manipulating matrices, in what seems to just be a complicated way to rewrite basic algebra. Most school teachers can't tell you why this is worthwhile, but matrices turn out to be extremely important in more advanced physics. So how can you tell what you need to know? In general, you can avoid this problem by sticking to good books. They'll contain exactly what actually matters.

Can I still learn physics if I'm behind? / Is it okay to skip ahead?

Sometimes young students give up because they know people younger than them who are "ahead." This is silly, because learning is about what *you* do, not what others do. Success in physics requires accumulating a body of knowledge, which takes on the order of one year for high school physics competitions, and ten for physics research. People who get to that point earlier in life just get a few extra years to use it; they don't stop you from doing the same.

[10] And learning these questionable rules is the good part; most of the time you just work on cutesy crafts, like posters, mosaics, and collages, while not being taught any actual information. These rituals are largely unique to America, and stem from educational philosophies which suggest that all true learning is constructed by students on their own. In practice, we spent ages giving presentations copy-pasted from Wikipedia, with the teacher occasionally encouraging us to "use critical thinking", which was kind of hard when nobody knew what the hell was going on. For any student who's serious about learning, the standard public school curriculum wastes multiple years by age 13.

[11] Other traditional wastes of time include writing cursive (which was only more efficient than print centuries ago, when people used quills and fountain pens) and manually typing MLA or APA citations. On the other hand, some teachers want to remove multiplication tables, but having facts like $4 \times 5 = 20$ memorized is actually extremely useful.

When I was in 10th grade, I found the USAPhO completely impossible. But the same year, one of my classmates made the US Physics Team. I found this encouraging rather than discouraging, since it meant you could succeed just by self-study. And over the next year, we both improved through separate study, and both won IPhO gold medals. As another example, after graduating from college, I took a fun two-year detour in the UK. When I finally started my PhD, I was "behind" almost everybody the same age as me, but "ahead" of almost everybody the same year as me. It's not worth obsessing over comparison, because it always depends on what you're comparing to.

Insecure adults discourage younger students from "skipping ahead", but this is silly as well. The all-time second top voted post on Reddit's AskPhysics is a warning to avoid learning anything faster than your high school teaches. But that doesn't even make sense, because different schools go at different paces. At a typical American public school, Algebra I is taught in 8th grade (age 13). But in many European and Asian countries, this material is covered earlier, while the California Math Framework says that all students should learn it together at age 14. So you should never fear being ahead of the "usual" track, because it's constantly changing, and set by politicians for reasons that have little to do with science. Sadly, some teachers discourage their students from learning more, because they find it more convenient if the whole class marches together. If this happens to you, I recommend seceding: quietly sit at the back of the class, and teach yourself whatever you want. Set your goals based on your knowledge and interests, not your age.

Sir, what are your tips to crack JEE?

The JEE is often called the world's hardest exam. Of the million students that take it every year, many of whom spend their teenage years living in coaching institutes, only about 1% are admitted to the IITs. But the JEE is a bad exam. It rewards speed and memorization, and it has many defective questions which can only be solved by guessing what the exam writer meant.

On the question-and-answer site Physics StackExchange, we saw dozens of cases where JEE students expressed their doubts about ambiguous questions, or complained about being made to memorize trivia that wasn't even true. (For some randomly chosen examples, see 1, 2, 3, 4, 5.) The JEE teaching material I've seen focuses on tossing out a vast array of magic formulas, without explaining where they come from or what they even mean. For instance, in the **E8** handout, I discuss how many JEE sources tell their students to remember the "magnetic length" of a bar magnet. Each source gives a different number, and none of them explain that this is an obsolete rule of thumb, which doesn't apply to modern rare-earth magnets.

Even the hardest questions on the JEE Advanced have issues. In one case, the exam included a beautiful, classic puzzle which is also included in **M4**. But this is silly exam design, because it's not reasonable to expect a student to solve such a question from scratch in 3 minutes. Instead, the question mostly rewards students who have already seen and memorized the answer. The arguably most famous JEE question in history is a 3D rotation problem from 2016. (There are dozens of YouTube videos where JEE students celebrate it stumping graduate students and even professors.) But as I discuss in **M8**, the problem isn't correctly posed. Under the standard definition of angular velocity, none of the answers were correct. You can only solve it by guessing what they meant.

Indian students often tell me they skip learning "theory" to save time for "problem solving". In other words, they plan to memorize a lot of formulas without asking why they work, or even if they make sense. But in such an environment, competitive exams degrade from a useful tool to motivate and measure learning, into an enormous waste of time. Even members of the Indian Olympiad team have complained to me that the system serves to discourage deeper thought. An exam isn't good just because it has a low pass rate, and I hope Indian students get something better soon.

Can physics competitions help me with top American college admissions?

The American elite college admissions system is one of the most dysfunctional, dehumanizing systems in the world. It's so bad that I think it's done noticeable damage to American society.

The problem is that most of the application carries little information. Due to rampant grade inflation at both public schools and private schools, every reasonably competitive student has a near-perfect GPA, and since COVID, many top schools stopped taking standardized test scores. You can list your achievements on your resume, but the admissions officers don't reliably know the difference between the Physics Olympiad, the much easier Physics Bowl, the much harder Physics Cup, or a random competition that popped up yesterday.[12] You can write about how much you love physics, but it won't land: they don't know what Maxwell's equations are, and may no longer remember the quadratic formula.[13] (And besides, anybody can use AI to say the same.)

So in practice, the decision rests on vibes. A random admissions officer you've never met will quickly skim your essay and try to judge your character and pedigree – largely by collapsing you into a stereotype. For instance, Harvard gives a "personality" rating to all candidates. In recent lawsuits, they were forced to disclose that Asian students get far lower ratings, while students whose parents have a top 1% income have far higher ratings. A recent Supreme Court decision made such behavior illegal, but in practice it's almost impossible to prove it's happening. Admissions statistics have hardly changed, and some admissions officers have even bragged about ignoring the law.

Even if they did follow the law, the real problem with the system is its air of moral judgment. If the main criterion is character, then being rejected is an implicit judgment of poor character. I've seen many cases where brilliant, kind, and charismatic students are rejected from all top colleges – but when other people hear about it, they simply assume it must have been because that student is a *bad person*, in some subtle way only an admissions officer can detect.

To avoid this dreaded outcome, tens of thousands of students do everything they can to look impressive. At "top" American high schools, students sign up for a dozen AP classes they barely care about, leading to huge loads of busywork that prevent them from learning anything else. Students create fake clubs which never meet, except to elect a gigantic slate of "leadership" positions and make a shiny website listing them. They even pay lawyers to found fake nonprofits in their name, and in some schools, dozens of such organizations are created per year. As I complain here, even science is corrupted, as students pay thousands to buy fake publications. I've seen numerous students try to pretend they're not Asian, by carefully scrubbing any hints from their apps. The humiliating finale is writing the personal statement, where students take the most painful, intimate moments of their lives, and force them into cliche, inspirational stories of personal growth.[14]

It's disturbing that the main path into America's elite rewards appearances and dishonesty from such a young age. When people take the A-levels in the UK, the Concours in France, the Abitur in Germany, the Gaokao in China, the CSAT in South Korea, or even the JEE in India, there's at least some objective target under the student's control. Students know that they can do better by learning specific skills. In America, students strive to look superficially inspiring to a random

[12] It's easy to forget this if you focus exclusively on math and physics, but students in general can specialize into *hundreds* of activities, such as quizbowl, painting, dance, and countless sports and musical instruments, each of which has dozens of similarly-named competitions that vary enormously in rigor. It's too much for an average admissions officer to keep track of. So if you mention your $F = ma$ exam score, they certainly won't know what counts as a high score, they probably won't know the exam is nontrivial, and they might not even realize that it's about physics!

[13] That's because the vast majority of admissions officers have degrees in humanities, not the sciences. After all, who else would enjoy reading a hundred essays per day?

[14] On an unrelated note, in the past decade the rate of student disabilities at Stanford has tripled. Now about 40% of students are officially registered as disabled, which yields additional time on exams, and better housing and food.

person they'll never meet, who can't check anything they're saying. The anxiety never ends, since there's no way to know or control the outcome, short of having parents who can donate millions.

This system leaves no room for real education, and the only dignified response is to ignore it. So if you only wanted to do physics for your college apps, this is your warning! It's not the One Weird Trick to getting in. Instead, physics competitions provide something even more important: an escape. They're one of the few places a teenager with no budget can work on nontrivial problems, to see if they really understand something. They give you an excuse to learn beautiful and universal truths, judged by nature rather than capricious people. It's knowledge that nobody can take away.

And I have some good news: STEM-focused schools such as MIT and Caltech understand the Ivy League's problems, and have implemented real fixes. They have specialized training programs for their admissions officers, which explain the differences between competitions. They even have dedicated offices to check whether high school research is legitimate, which draws on the expertise of actual professors. MIT in particular is a paragon of transparency and rigor. In 2023, it was the only top US school that required SAT/ACT scores, and it remains the only one that doesn't give legacy preference. Compared to every Ivy League school, it has less wealthy incoming students, but its outgoing students become the most wealthy a decade later. Its admissions officers care deeply about doing their jobs well, and even write public blog posts about it. Most US Physics Team finalists (attendees of the training camp) are admitted to MIT and choose to go there.

Do I have enough talent to succeed in physics?

In response to this difficult question, many well-intentioned adults assert that talent does not exist, or that anybody can do anything if they really try. These trite sentiments come from a good place, but they're rarely satisfying to their recipients because they're clearly not true. Talent does exist. It's the reason that wealthy families can spend tens of thousands of dollars propping up or outright falsifying their children's SAT scores, to get outscored by less advantaged kids using only the $20 Blue Book. More dramatically, it's the reason that Ramanujan went from being the son of an Indian clerk, doing mathematics alone in near starvation, to the apex of mathematics in Cambridge.

When I was a kid, I naively thought every "level" of physics required some minimum bar of talent, and that people just kept climbing until they hit a level of abstraction they were simply unable to grasp. After all, adults would say things like, "math stopped making sense for me at trigonometry," or "I couldn't make it past differential equations." So when things got hard, such as when I started quantum field theory, I had a sinking feeling that I was "hitting the wall."

But in reality, the cognitive load of learning stays relatively constant. With modern resources, the difficulty of learning quantum mechanics is about the same as learning introductory physics, *provided you have equal mastery of the prerequisites.* The reason people hit walls is largely not because the material gets inherently harder, but because they finally fall through the massive holes in their foundations. That's the problem; it can't be the raw complexity of manipulating the symbols, because all of us can follow a much larger set of rules for manipulating a much larger set of symbols, whenever we assemble letters into words and sentences.

So let's say that when you first learned physics, things clicked for you. You saw the world in a different way, it felt good, and you wanted to learn more. If this rings true, I can assure you that if you keep going, it'll keep paying off. You'll continue to get "aha!" moments. You'll continue to piece together, with concentrated effort, new ways of looking at the world. Of course, the *rate* at which you do this depends on talent. But if you've made physical insights before, you will make them in the future, if your foundations are good. There is no wall; how far you go is up to you.